

# Sum and direct sum

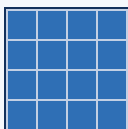
One construction with one condition attached. Whether the intersection is trivial decides the dimension formula, the uniqueness of decompositions, and whether the notation  $\oplus$  is permitted at all.

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CASES

## THE GENERAL CASE

1



**Sum** |  $W + W = \{w + w\}$

**DIMENSION**

$$\dim W + \dim W - \dim(W \cap W) = 2 + 2 - 1 = 3$$

**INTERSECTION**

may be nontrivial

**DECOMPOSITION**

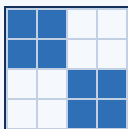
exists, but not unique

The correction term is a double count: anything in both subspaces is added twice and must be removed once. Two planes through the origin in  $\mathbb{R}^3$  have dimensions summing to four in a three-dimensional space — the intersection line is the missing one.

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## WHEN THE INTERSECTION IS TRIVIAL

1



**Direct sum** |  $W \cap W = \{0\}$

**DIMENSION**

$$\dim W + \dim W = 2 + 2 = 4$$

**INTERSECTION**

trivial — the origin only

**DECOMPOSITION**

exists and is unique

With nothing shared there is nothing to subtract, and each vector splits in exactly one way. This is the structure behind **orthogonal complements** and behind every basis:  $V$  is the direct sum of the lines spanned by its basis vectors.

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